

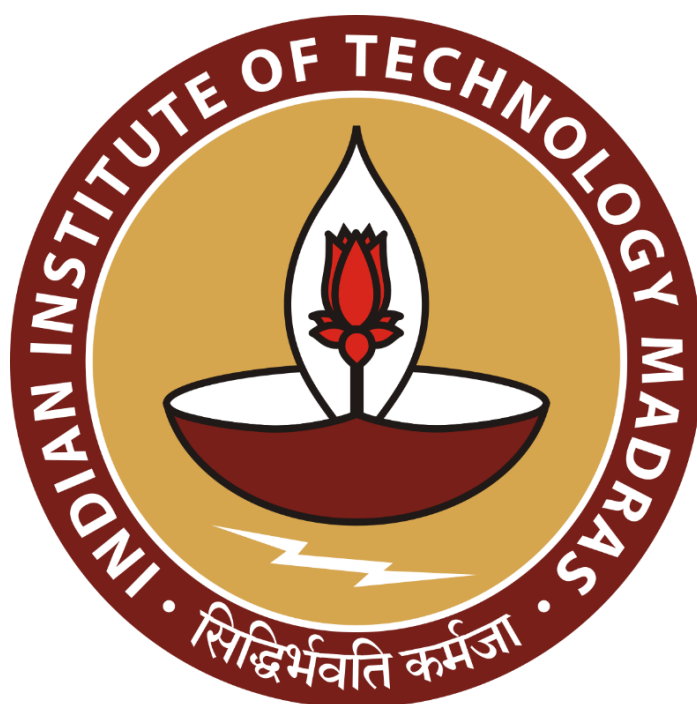
DATA-DRIVEN ANALYSIS OF POULTRY FARM OPERATIONS AND LIVE CHICKEN SUPPLY

A Proposal report for the BDM capstone Project

Submitted by

Name: PULLURI YASHWANTH KRISHNA

Roll Number: 23f1002174



IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

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Declaration Statement

I am working on a Project titled “**DATA-DRIVEN ANALYSIS OF POULTRY FARM OPERATIONS AND LIVE CHICKEN SUPPLY**”. I extend my appreciation to **Karthik Reddy Poultry**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

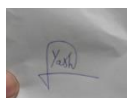
Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate:



Name: Pulluri Yashwanth Krishna

Date: 13/10/2025

1 Executive Summary

The project is focused on studying and providing recommendations for the challenges faced by the business Karthik Reddy Poultry, a small poultry farm located in Chinna Masanpally village, Rayapole Mandal, Siddipet District, Telangana. The farm mainly works as a B2B business, supplying live chickens to local meat shops, hotels, restaurants and other regional businesses.

The major challenges at Karthik Reddy Poultry lie in its day-to-day operations and management. These include frequent chick deaths, inconsistent vaccine and medicine tracking, and inefficient waste Handling. These issues have made it harder for the poultry farm to run smoothly, plan properly and keep profits steady.

To get a better understanding, I collected data from the farm and looked closely how things are done. Data was collected from farm through observing activities, talking with staff, and reviewing records. For analysis on data, I will use Excel, Python, Machine Learning and data visualization for simple dashboards. The analyses will focus on identifying performance gaps and suggesting actionable improvements in production, vaccination management, and waste management.

The goal of the project is to provide suggestions to farm that help reduce chick deaths, make the farm more productive, keep vaccinations on track, and improve the overall profitability for Karthik Reddy Poultry. The Project outcomes will help farm to manage resources effectively.

2 Organisation Background

Business Name: Karthik Reddy Poultry

Address: Chinna Masanpally, Rayapole, Siddipet, Telangana, 502278

Owner's Name: Mr. Karthik Reddy

Karthik Reddy Poultry is a small-scale poultry farm that mainly runs as B2B business, that supply live chickens to local meat shops, hotels, restaurants, and other businesses in the region. The poultry farm was founded in 2023 by Mr. Karthik Reddy, who got idea from his father, Mr. Laxma Reddy, who is both a farmer and a businessman managing agricultural and borewell services. The Poultry farm maintains multiple poultry batches throughout the year, focusing on quality, hygiene and timely supply.

3 Problem Statements

- 3.1 **High Chick Mortality:** The poultry farm faces major issue of frequent chick deaths mainly due to poor hygiene, lack of proper disease monitoring, temperature control, irregular vaccination and other factors. Sometimes pathogenic infections like bird flu can cause major losses.
- 3.2 **Inconsistent Vaccine and Medicine Tracking:** There is no proper system to track vaccines and medicines. This affects bird health and increases costs, highlighting the requirement of proper tracking record. For optimized batch scheduling, we need consistent tracking of vaccination.
- 3.3 **Poor Waste Management:** Wastes such as dead birds not handled properly causes bad odor and hygiene issues. Efficient waste management is essential for the growth of the farm. Solid waste management is essential for reduction of infections to chickens in the farm and enhance Hygiene in the farm.

4 Background of the Problem

Karthik Reddy Poultry faces a major issue that is frequent chick mortality. This is a small-scale Poultry business so operational issues are common. Chick death can directly affect productivity and profitability of the farm. Chick deaths are common due to inadequate disease monitoring, irregular vaccination, poor hygiene and improper temperature control. This chick mortality can severely impact farm revenue and can constitute to losses.

Secondly, the Farm is facing inconsistent tracking of vaccines and medicines. There is a need of proper system to monitor vaccination schedules and medicine usage for bird health. For optimized batch scheduling, we need vaccination tracking. Generally, a batch will be 45-55 days. If a bird Health is poor, period can be extended. This can impact supply of live chickens and can show impact on Revenue. Health monitoring System is essential for the chick overall growth. This ensures timely treatment and also can reduce chick mortality.

Poor Waste management is an additional challenge. Improper disposal of dead birds and other farm waste leads to unhygienic conditions. This can lead to infections among birds and sometimes can affect workers in the farm. Solid Waste Management is required to obtain clean conditions in the farm. Alternatives are needed for managing of waste.

5 Problem Solving Approach

5a. Details about the methods used with Justification:

Given the nature of the problems faced by 'Karthik Reddy Poultry,' a comprehensive approach involving both quantitative and qualitative methods is required.

Quantitative Methods:

Time-Series Analysis: Chick mortality, vaccination schedules, and production trends will be analyzed over time to and production over time. This Analysis will Help identify trends, seasonal patterns, and periods of higher mortality, allowing interventions. This Analysis will idea how production and quantity varies from batch to another batch.

Statistical Computation: Descriptive statistics (mean, median, standard deviation) will analyze mortality rates, vaccine coverage, and waste generation and this will handle inefficiencies.

Qualitative Methods:

Talking with Business Owner: Conversations with farm staff and the owner provides insights in hygiene practices, vaccine administration, and waste disposal routines.

Benchmarking: Comparing with other farm's practices in region provides their practices for vaccination tracking, disease control, and waste management.

5b. Details about the intended data collection with Justification

Chick Mortality data: collected mortality data from farm batches on past batches. These include Variables like Daily/weekly deaths, causes, age of chicks, temperature, humidity, hygiene score.

Vaccination and Medicines: collected medical and vaccination data from records. These include variables like Dates, batch IDs, type of vaccine/ medicine, dosage, administrator, Status, next scheduled date.

Waste Management: Collected data regarding Waste in the farm. The variables that include the waste management are Waste type, quantity, disposal method, frequency, hygiene score.

Justification:

Collecting these variables ensures a direct link to the problems. Mortality and vaccination data highlight health risks. Waste and environmental data reveal hygiene gaps; feed and ventilation data help to correlate operational practices with bird health. These insights guide better outcomes.

5c. Analysis Tools and Justification:**Analysis Tools:**

Excel / Google Spreadsheets: These tools are used for initial stages of data handling and data cleaning. They include all raw data related to chick batches, mortality, vaccination records, and waste management will be compiled and organized in clear format. Calculations such as mortality rates, vaccination tracking and waste generation metrics will be carried out using Spreadsheets formulas.

Python (Pandas, Matplotlib, Seaborn): Python is used for deeper analysis, correlations, trend visualization, and prediction of high-risk periods for mortality. Libraries like matplotlib and seaborn will be used to generate heatmaps, line plots and correlation graphs, enabling identification of underlying patterns such as temperature fluctuations etc.

Complementing Excel's capabilities for a more comprehensive understanding of the dataset.

Justification:

Excel and Spreadsheets provide easy-to-use for basic computation and trend visualization.

Python enables deeper analysis, pattern recognition, and predictive modeling.

Python enables advanced data manipulation and sophisticated visualizations, Complementing Excel's capabilities for a more comprehensive understanding of the dataset. The tools will provide evidence-based decisions.

6 Expected Timeline

6.1 Work Breakdown Structure:

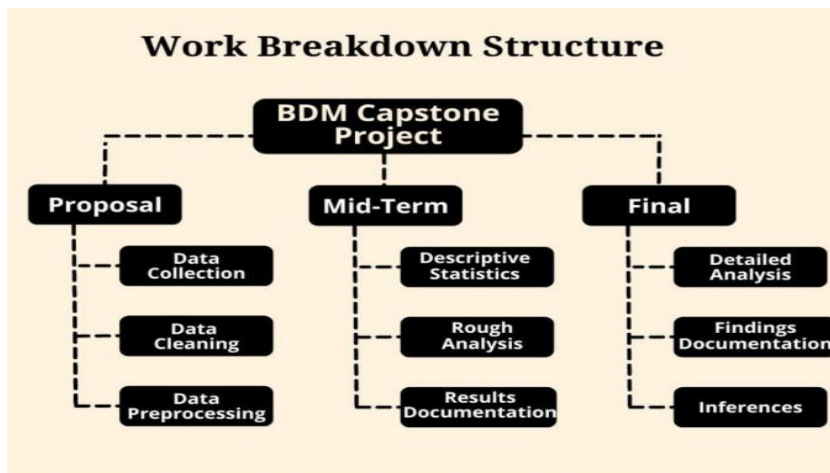


Image 2 : Work Breakdown Structure

6.2 Gantt chart

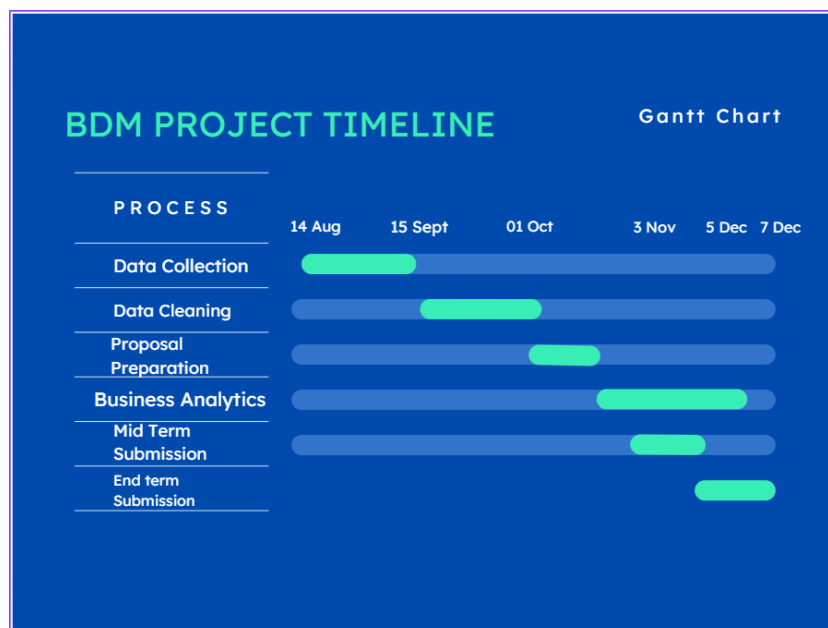


Figure 1 Expected timeline for completion of project.

7 Expected Outcome

To have a clear understanding of factors affecting chick mortality through data analysis on hygiene, temperature, vaccination, enabling reduction in death rates of chicks.

- To establish an effective vaccination and medicine tracking system that ensures timely administration, accurate batch scheduling and profits. To assist poultry owner in efficient waste managing & data drive decisions.